based on P335

(Roy Glauber, Nobel 2005)

Coherent states of the harmonic oscillator (a.k.a. Glauber states in quantum optics)

$$\hat{a}_- \varphi_\alpha(x) = \alpha \varphi_\alpha(x)$$

$$\hat{a}_- = \frac{1}{\sqrt{2\hbar m\omega}}\left(i\hat{p} + m\omega\hat{x}\right) = \frac{1}{\sqrt{2\hbar m\omega}}\left(\hbar\frac{d}{dx} + m\omega x\right)$$

Use (eq. 2.64): $\int_{-\infty}^{+\infty} f^*(a_\pm g)\,dx = \int_{-\infty}^{+\infty}(a_\mp f)^* g\,dx$

($\hat{a}_\mp$ = hermitian conjugate of $\hat{a}_\pm$)

Let us just suppose that $\varphi_\alpha(x)$ exists.

(a) $$\langle x\rangle_\alpha = \sqrt{\frac{\hbar}{2m\omega}}\int \varphi_\alpha^*(\hat{a}_+ + \hat{a}_-)\varphi_\alpha\,dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\int\left[(\hat{a}_-\varphi_\alpha)^*\varphi_\alpha + \varphi_\alpha^*\alpha\varphi_\alpha\right]dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}}(\alpha^* + \alpha) = \sqrt{\frac{\hbar}{2m\omega}}\,2\,\mathrm{Re}(\alpha).$$

$$\langle p\rangle = i\sqrt{\frac{\hbar m\omega}{2}}\int \varphi^*(\hat{a}_+ - \hat{a}_-)\varphi\,dx =$$

$\alpha = \alpha' + i\alpha''$

$\alpha = \mathrm{Re} + i\,\mathrm{Im}$

$$= i\sqrt{\frac{\hbar m\omega}{2}}\int\left((\hat{a}_-\varphi)^*\varphi - \varphi^*\alpha\varphi\right)dx \qquad (\alpha^* - \alpha)$$

$$= \sqrt{\frac{\hbar m\omega}{2}}\,2\,\mathrm{Im}(\alpha) = i\sqrt{\frac{\hbar m\omega}{2}}(\alpha^* - \alpha).$$

$$\langle x^2\rangle = \frac{\hbar}{2m\omega}\int \varphi^*(\hat{a}_+ + \hat{a}_-)(\hat{a}_+ + \hat{a}_-)\varphi\,dx$$

$$= \frac{\hbar}{2m\omega}\int \varphi^*\left(\hat{a}_+\hat{a}_+ + \hat{a}_-\hat{a}_- + \hat{a}_+\hat{a}_- + \underbrace{\hat{a}_-\hat{a}_+}_{1+\hat{a}_+\hat{a}_-}\right)\varphi\,dx$$

$$= \frac{\hbar}{2m\omega}\int\left\{(\hat{a}_-\hat{a}_-\varphi)^*\varphi + \alpha^2\varphi^*\varphi + 2(\hat{a}_-\varphi)^*\alpha\varphi + \varphi^*\varphi\right\}dx$$

$$= \frac{\hbar}{2m\omega}\left((\alpha^*)^2 + \alpha^2 + 2\alpha^*\alpha + 1\right) = \frac{\hbar}{2m\omega}\left\{(\alpha^* + \alpha)^2 + 1\right\} = \langle x\rangle^2 + \frac{\hbar}{2m\omega}$$

$$\langle p^2 \rangle = -\left(\frac{\hbar m\omega}{2}\right) \int \varphi^* (\hat{a}_+ - \hat{a}_-)(\hat{a}_+ - \hat{a}_-) \varphi \, dx$$

$$= -\frac{\hbar m\omega}{2} \int \varphi^* \left(\hat{a}_+\hat{a}_+ + \hat{a}_-\hat{a}_- - \hat{a}_+\hat{a}_- - \underbrace{\hat{a}_-\hat{a}_+}_{1+\hat{a}_+\hat{a}_-}\right) \varphi \, dx$$

$$= -\frac{\hbar m\omega}{2} \int \left\{ (\hat{a}_-\hat{a}_-\varphi)^* \varphi + \varphi^* \alpha^2 \varphi - 2(\hat{a}_-\varphi)^* \alpha \varphi - |\varphi|^2 \right\} dx$$

$$= -\frac{\hbar m\omega}{2} \left(\alpha^{*2} + \alpha^2 - 2\alpha^*\alpha - 1\right) = \frac{\hbar m\omega}{2} + \langle p \rangle^2.$$

(b) $\Rightarrow \sigma_p^2 = \langle p^2 \rangle - \langle p \rangle^2 = \frac{\hbar m\omega}{2}$

$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{\hbar}{2m\omega}$

$\sigma_x^2 \sigma_p^2 = \frac{\hbar^2}{4}$

$\sigma_x \sigma_p = \frac{\hbar}{2}$ for any value of α.

(c) $\varphi_\alpha = \sum_{n=0}^{\infty} c_n \psi_n$ in terms of w.f. of harmonic oscillator.

Note $\psi_n = \frac{(\hat{a}_+)^n}{\sqrt{n!}} \psi_0$.

Find c_m: $\int \psi_m^* \varphi_\alpha \, dx = \sum_n c_n \int \psi_m^* \psi_n \, dx = c_m$

On the other hand $\int \psi_m^* \varphi_\alpha \, dx =$

$$= \int \left(\frac{(\hat{a}_+)^m}{\sqrt{m!}} \psi_0\right)^* \varphi_\alpha \, dx = \frac{1}{\sqrt{m!}} \int \psi_0^* \underbrace{\left((\hat{a}_-)^m \varphi_\alpha\right)}_{\alpha^m \varphi} dx$$

$$= \frac{\alpha^m}{\sqrt{m!}} \underbrace{\int_{-\infty}^{\infty} \psi_0^* \varphi_\alpha \, dx}_{c_0} \Rightarrow c_m = \frac{\alpha^m}{\sqrt{m!}} c_0$$

$$\Rightarrow \varphi_\alpha(x) = c_0 \sum_n \frac{\alpha^n}{\sqrt{n!}} \psi_n(x)$$

(d) Find c_0 by normalizing:

$$1 = \int_{-\infty}^{\infty} dx\, |\varphi_\alpha|^2 = |c_0|^2 \sum_{nm} \frac{(\alpha^*)^m \alpha^n}{\sqrt{m!\,n!}} \underbrace{\int \psi_m^* \psi_n \, dx}_{\delta_{mn}}$$

$$= |c_0|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} = |c_0|^2 e^{-|\alpha|^2}$$

$$\Rightarrow c_0 = e^{-|\alpha|^2/2}.$$

$$\varphi_\alpha(x) = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \psi_n(x)$$

(e) let $\varphi_\alpha(x) = \varphi_\alpha(x,0)$ be the initial state.

$$\varphi_\alpha(x,t) = \sum_n c_n e^{-iE_n t/\hbar} \psi_n(x) = \left(E_n = \hbar\omega\left(n+\tfrac{1}{2}\right)\right) =$$

$$= e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} e^{-i\frac{\hbar\omega}{\hbar}t\left(n+\frac{1}{2}\right)} \psi_n =$$

$$= e^{-|\alpha|^2/2 - i\frac{\omega t}{2}} \sum_n \frac{\left(\alpha e^{-i\omega t}\right)^n}{\sqrt{n!}} \psi_n$$

Thus $$\hat{a}_- \varphi_\alpha(x,t) = e^{-\frac{|\alpha|^2}{2} - i\frac{\omega t}{2}} \sum_{n=0}^{\infty} \frac{\left(\alpha e^{-i\omega t}\right)^n}{\sqrt{n!}} \underbrace{\hat{a}_- \psi_n}_{\sqrt{n}\,\psi_{n-1} \text{ for } n\geq 1}$$

$$= e^{-\frac{|\alpha|^2}{2} - i\frac{\omega t}{2}} \sum_{n=1}^{\infty} \frac{\left(\alpha e^{-i\omega t}\right)^n}{\sqrt{(n-1)!}} \psi_{n-1} =$$

$$= \left(\alpha e^{-i\omega t}\right) e^{-\frac{|\alpha|^2}{2} - i\frac{\omega t}{2}} \sum_{n=0}^{\infty} \frac{\left(\alpha e^{-i\omega t}\right)^n}{\sqrt{n!}} \psi_n = \alpha e^{-i\omega t} \varphi_\alpha(x,t)$$

$\Rightarrow \varphi_\alpha(x,t)$ is coherent state for any t!

$$\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\alpha e^{-i\omega t} + \alpha^* e^{i\omega t}\right) = \sqrt{\frac{\hbar}{2m\omega}}\, 2|\alpha| \cos(\theta + \omega t)$$

$$\alpha = |\alpha| e^{i\theta}$$

$$\langle p\rangle = i\sqrt{\frac{\hbar m\omega}{2}}\left(\alpha^* e^{i\omega t} - \alpha e^{-i\omega t}\right)$$

$$|\alpha|\left(e^{i(\theta+\omega t)} - e^{-i(\theta+\omega t)}\right) = 2i\,|\alpha|\,\sin(\theta+\omega t)$$

$$\langle p\rangle = -\sqrt{\frac{\hbar m\omega}{2}}\;2|\alpha|\,\sin(\theta+\omega t).$$

Note that $\langle p\rangle = m\frac{d\langle x\rangle}{dt} = m\frac{d}{dt}\sqrt{\frac{\hbar}{2m\omega}}\;2|\alpha|\cos(\theta+\omega t)$

Just as Classical pendulum: $\begin{cases} x = x_0\cos(\theta+\omega t) \\ p = p_0\sin(\theta+\omega t)\end{cases}$

~~Note: $\frac{\langle p\rangle^2}{2m} + \frac{1}{2}m\omega^2\langle x^2\rangle$~~

Coherent states describe <u>almost</u> classical oscillating motion

with Energy: $E = \frac{\langle p\rangle^2}{2m} + \frac{1}{2}m\omega^2\langle x^2\rangle =$

$$= \frac{1}{2m}\left(\langle p\rangle^2 + \frac{\hbar m\omega}{2}\right) + \frac{1}{2}m\omega^2\left(\langle x\rangle^2 + \frac{\hbar}{2m\omega}\right) =$$

$$= \frac{\hbar\omega}{4} - \frac{1}{2m}\frac{\hbar m\omega}{2}(\alpha^*-\alpha)^2 + \frac{1}{2}m\omega^2\frac{\hbar}{2m\omega}(\alpha^*+\alpha)^2 + \frac{\hbar\omega}{4} =$$

$$= \frac{\hbar\omega}{2} - \frac{\hbar\omega}{4}(\alpha^*-\alpha)^2 + \frac{\hbar\omega}{4}(\alpha^*+\alpha)^2 =$$

$$= \frac{\hbar\omega}{4}\left[\alpha^{*2} + 2|\alpha|^2 + \alpha^2 - (\alpha^{*2} - 2|\alpha|^2 + \alpha^2)\right] + \frac{\hbar\omega}{2}$$

$$= \hbar\omega\left(|\alpha|^2 + \frac{1}{2}\right) = \text{const even when } \alpha = \alpha e^{-i\omega t}!$$

(f) Note that $\hat{a}_-\varphi_{\alpha=0}(x) = 0 \Rightarrow \varphi_0(x)$ is a coherent state with $\alpha=0$.

Finally: $\varphi_\alpha(x) = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\left(\frac{(\hat{a}_+)^n}{\sqrt{n!}}\varphi_0\right) = e^{-\frac{|\alpha|^2}{2}}\sum_n\frac{(\alpha\hat{a}_+)^n}{n!}\varphi_0 =$

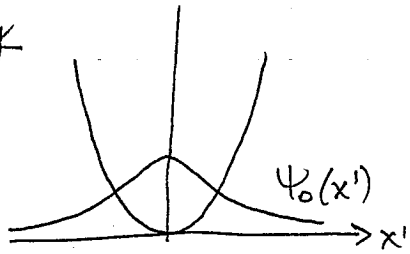
$$= e^{-\frac{|\alpha|^2}{2}}e^{\alpha\hat{a}_+}\varphi_0 \quad !$$

Let us now think of the following problem:

[Charged particle is in the ground state of harmonic oscillator.
Suddenly, we turn on a uniform electric field E.
Describe time evolution of the particle.

<u>Initially</u>

$(t<0)$



$$P.E. = \frac{1}{2} m\omega^2 x'^2$$

I use x' for coordinate for later convenience

<u>At $t=0$ and later:</u>

$$P.E. = \frac{1}{2} m\omega^2 x'^2 - qEx'$$

potential energy of charge q in electric field E.

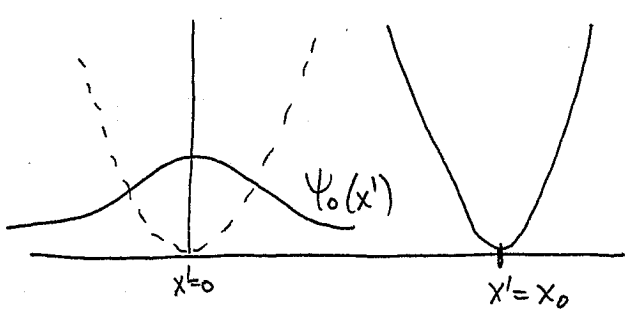
But note that $P.E. = \frac{1}{2} m\omega^2 x'^2 - qEx' =$

$$= \frac{1}{2} m\omega^2 (x'-x_0)^2 - \frac{1}{2} m\omega^2 x_0^2$$

where $x_0 = \dfrac{qE}{m\omega^2}$

Thus new potential energy is again that of harmonic oscillator but located at $x_0 = \dfrac{qE}{m\omega^2}$!

Denote $x = x' - x_0$



In the x-coordinate system, S.E. reads just

$$\hat{H}\psi = E\psi, \quad \hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2} m\omega^2 x^2 .$$

Initial state is $\psi_0(x') = \psi_0(x+x_0)$

Find: $\psi_0(x+x_0, t) = \sum_{n=0}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}$

That is, we need to express initial state $\psi_0(x+x_0)$ in terms of wave functions of harmonic oscillator $\psi_n(x)$ (centered about $x=0$, which means* $x'=x_0$).

We do this by connecting the problem to <u>coherent states</u>.

Observe that $\psi_0(x+x_0) = \psi_0(x) + x_0 \frac{d}{dx}\psi_0(x) + \frac{1}{2!} x_0^2 \frac{d^2}{dx^2}\psi_0(x)$

$$+ \dots + \frac{1}{n!} x_0^n \frac{d^n}{dx^n}\psi_0(x) =$$

$$= \exp\left(x_0 \frac{d}{dx}\right)\psi_0(x) = \underbrace{\hat{T}_{x_0}}_{\text{translation operator}} \psi_0(x)$$

But $\hat{p} = \frac{\hbar}{i}\frac{d}{dx} \Rightarrow \frac{d}{dx} = \frac{i}{\hbar}\hat{p}$

$$\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a}_+ - \hat{a}_-) \Rightarrow \frac{d}{dx} = -\sqrt{\frac{m\omega}{2\hbar}}(\hat{a}_+ - \hat{a}_-).$$

Thus $\psi_0(x+x_0) = \exp\left(-x_0\sqrt{\frac{m\omega}{2\hbar}}(\hat{a}_+ - \hat{a}_-)\right)\psi_0(x)$

For brevity, denote $\beta = x_0\sqrt{\frac{m\omega}{2\hbar}}$ (this is just dimensionless displacement.)

In <u>Appendix</u> it is proven that

$$e^{-\beta(\hat{a}_+ - \hat{a}_-)} = e^{-\frac{1}{2}\beta^2} e^{-\beta\hat{a}_+} e^{\beta\hat{a}_-}.$$

Next observe that $e^{\beta\hat{a}_-}\psi_0(x) = \left(1 + \beta\hat{a}_- + \frac{\beta^2}{2!}\hat{a}_-^2 + \dots + \frac{\beta^n}{n!}\hat{a}_-^n\right)\cdot\psi_0(x) = \psi_0(x)$ because all terms except 1 annihilate ψ_0: $\underline{\hat{a}_-\psi_0 = 0}$

Thus $\psi_0(x+x_0) = e^{-\frac{1}{2}\beta^2} e^{-\beta\hat{a}_+}\psi_0(x)$ $\left(\beta = x_0\sqrt{\frac{m\omega}{2\hbar}}\right)$

Compare with coherent state $\varphi_\alpha(x) = e^{-\frac{1}{2}|\alpha|^2} e^{\alpha\hat{a}_+}\psi_0(x)$

$$\Rightarrow \psi_0(x+x_0) = \varphi_{\alpha=-\beta}(x) \quad \left(\alpha = -x_0\sqrt{\frac{m\omega}{2\hbar}}\right)$$

Since $\varphi_\alpha(x,t) = e^{-\frac{1}{2}|\alpha|^2} e^{\alpha e^{-i\omega t} \hat{a}_+} \psi_0(x)$

we immediately conclude that

$$\psi_0(x+x_0,t) = e^{-\frac{1}{2}\left(x_0\sqrt{\frac{m\omega}{2\hbar}}\right)^2} e^{-x_0\sqrt{\frac{m\omega}{2\hbar}} e^{-i\omega t} \hat{a}_+} \psi_0(x)$$

Hence $\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}}\, 2\alpha \cos\omega t =$

$$= -\sqrt{\frac{\hbar}{2m\omega}}\, 2 x_0 \sqrt{\frac{m\omega}{2\hbar}} \cos\omega t = -x_0 \cos(\omega t).$$

$$\langle p \rangle = -\sqrt{\frac{\hbar m\omega}{2}}\, 2\alpha \sin(\omega t) =$$

$$= +\sqrt{\frac{\hbar m\omega}{2}}\, 2 x_0 \sqrt{\frac{m\omega}{2\hbar}} \cdot \sin(\omega t) = x_0 m\omega \sin(\omega t)$$

$$= m \frac{d}{dt} \langle x \rangle.$$

Expectation values $\langle x \rangle$ and $\langle p \rangle$ follow motion of the classical oscillator with initial displacement $\langle x(t=0) \rangle = -x_0$ and zero momentum $\langle p(t=0) \rangle = 0$.

Appendix Proof of Baker-Hausdorff formulae

For any two operators $\hat{A}$ and $\hat{B}$ such that $[\hat{A},\hat{B}] = C$ is the number,

the following identity holds:

$$e^{A} e^{B} = e^{A+B} e^{\frac{1}{2}[A,B]}.$$

Proof Consider $f(x) = e^{x\hat{A}} e^{x\hat{B}}$ and x is just a parameter.

$$\frac{df}{dx} = e^{xA} A e^{xB} + e^{xA} e^{xB} B = e^{xA} e^{xB} \left(e^{-xB} A e^{xB} + B\right)$$

$$= f(x)\left(e^{-xB} A e^{xB} + B\right)$$

$$\Rightarrow \quad \frac{1}{f}\frac{df}{dx} = e^{-xB} A e^{xB} + B. \qquad (1)$$

Next consider $g(x) = e^{-xB} A e^{xB}$:

$$\frac{dg(x)}{dx} = e^{-xB}(-BA)e^{xB} + e^{-xB} AB e^{xB} =$$

$$= e^{-xB}(AB - BA)e^{xB} = e^{-xB} \underbrace{C}_{\text{number!}} e^{xB} = e^{-xB} e^{xB} C = C.$$

thus $\frac{dg(x)}{dx} = C \Rightarrow g(x) = D + Cx$

but $g(0) = D = \hat{A} \Rightarrow D = \hat{A}$.

$$\Rightarrow g(x) = \hat{A} + Cx = \hat{A} + x[\hat{A},\hat{B}].$$

Back to (1) $\Rightarrow \frac{1}{f(x)}\frac{df(x)}{dx} = \hat{A} + [\hat{A},\hat{B}]x + \hat{B}$

$$\Rightarrow \frac{f(x)}{f(0)} = \exp\left[(\hat{A}+\hat{B})x + C\frac{x^2}{2}\right].$$

Since $f(0) = 1 \Rightarrow f(x) = e^{x(\hat{A}+\hat{B})} e^{\frac{1}{2}x^2 C}$

$$\Rightarrow f(x=1) = e^{\hat{A}} e^{\hat{B}} = e^{\hat{A}+\hat{B}} e^{\frac{1}{2}[\hat{A},\hat{B}]} \qquad \text{QED.}$$

Another way to express this:

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{2}[\hat{A},\hat{B}]} \quad \text{when } [\hat{A},\hat{B}] = c = \text{number}$$

We need $e^{-\beta(\hat{a}_+ - \hat{a}_-)} \Rightarrow \hat{A} = -\beta \hat{a}_+ \;,\; \hat{B} = \beta \hat{a}_-$

Then $e^{-\beta \hat{a}_+ + \beta \hat{a}_-} = e^{-\beta \hat{a}_+} e^{\beta \hat{a}_-} e^{-\frac{1}{2}[-\beta \hat{a}_+, \beta \hat{a}_-]} =$

$$= e^{-\beta \hat{a}_+} e^{\beta \hat{a}_-} e^{\frac{\beta^2}{2}[\hat{a}_+, \hat{a}_-]}$$

But $[\hat{a}_+, \hat{a}_-] = \hat{a}_+ \hat{a}_- - \hat{a}_- \hat{a}_+ = -1$

$\Rightarrow$ finally $e^{-\beta \hat{a}_+ + \beta \hat{a}_-} = e^{-\frac{1}{2}\beta^2} e^{-\beta \hat{a}_+} e^{\beta \hat{a}_-}$.